

UTM TENSILE REPORT — UTM_Test_20260817_103930_100%infill_Videocapture_2

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A₀) · anchor self-calibration

46.2 MPa

UTS

41.7 MPa

Yield σ_y

2.98 GPa

Modulus E

4.2 %

Failure ϵ_f

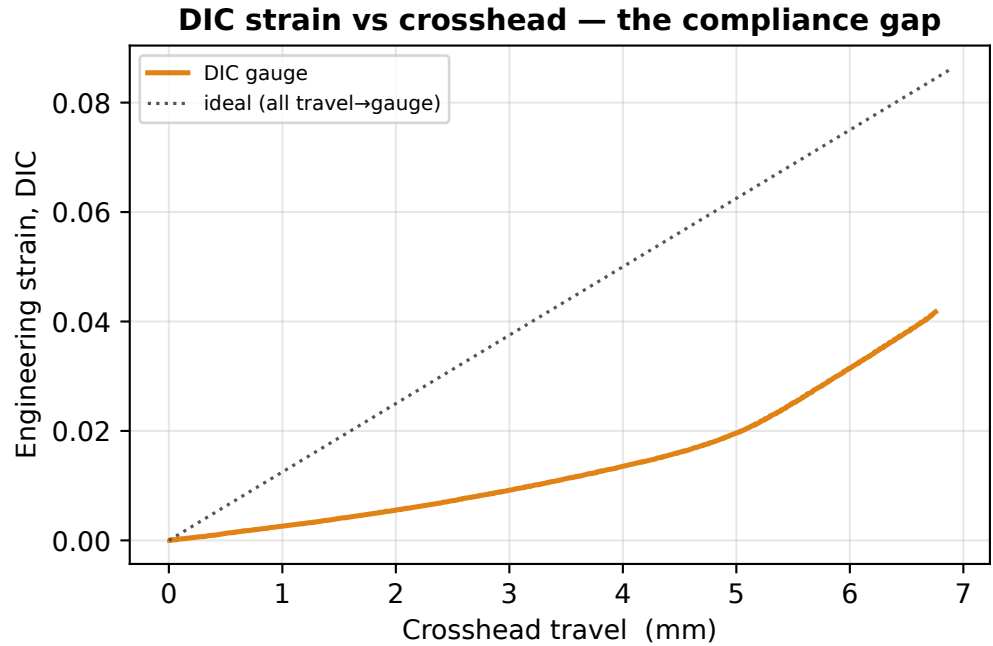
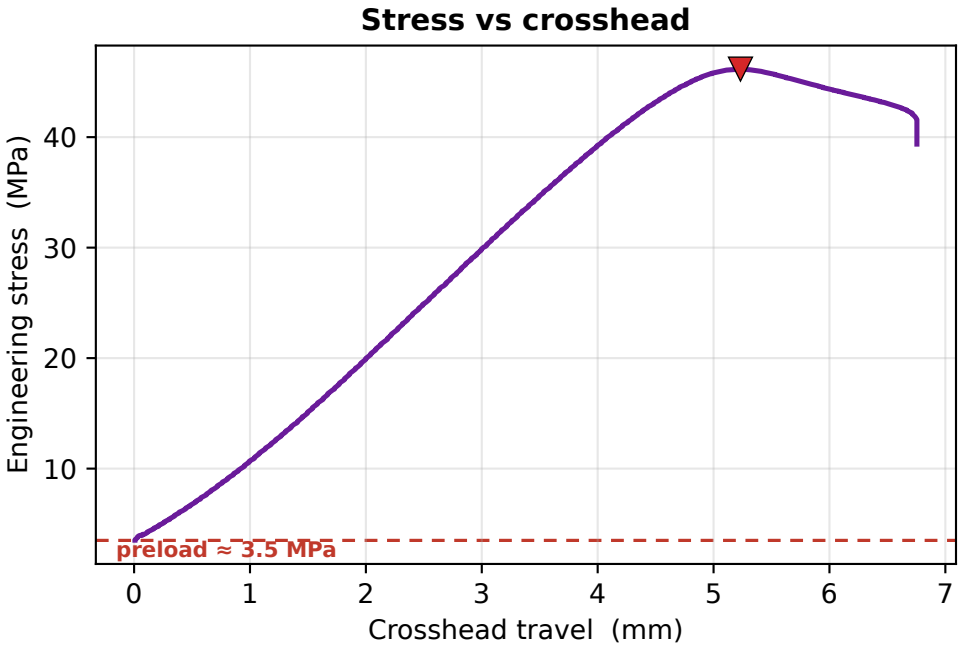
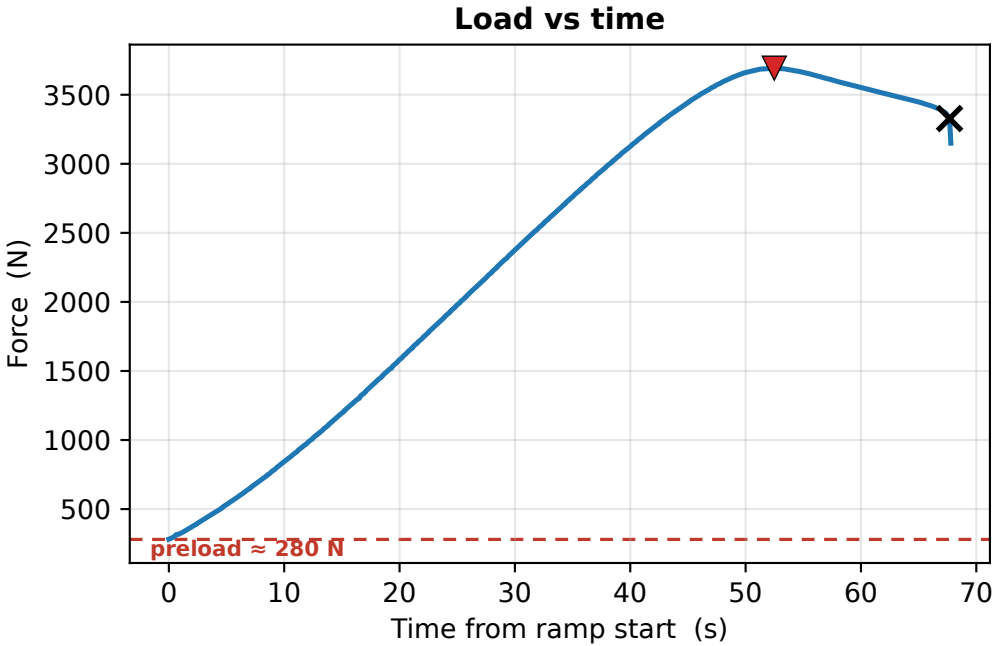
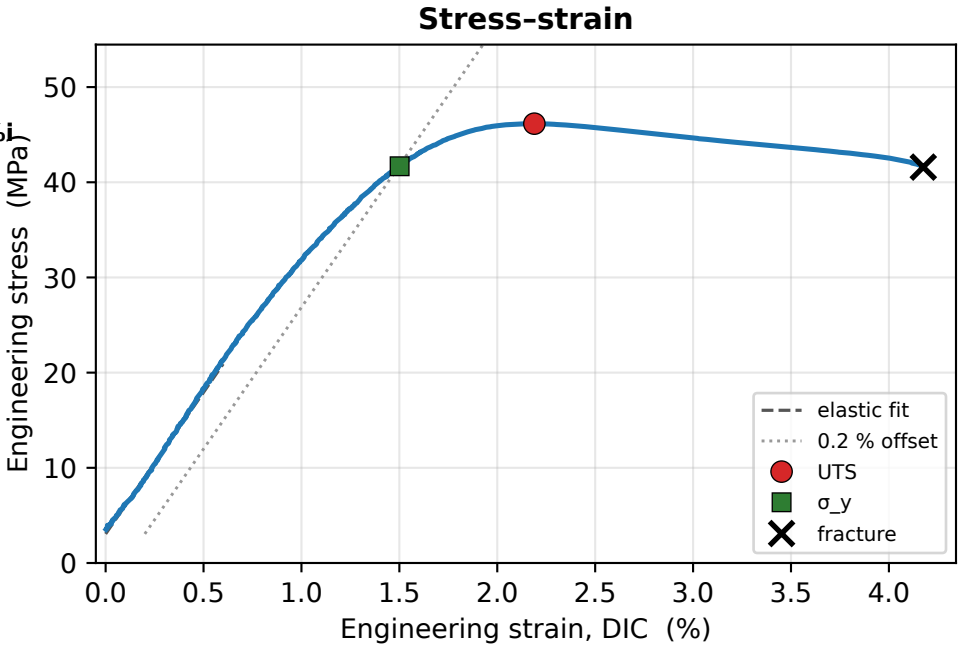
1564 kJ/m³

Toughness

280 N

Preload anchor

TEST SETTINGS	
Specimen	UTM_Test_20260817_103930_100%infill
DIC mode	White
Preload target	300 N
Test speed	0.100 mm/s
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	100 %
Force calib	sc -0.0065 / off 304.772
Comment	—
Test date	2026-08-17 10:37:38
Duration	145.3 s



VALIDATION vs REFERENCES			
prop	meas	Chacón	E-PLA k
UTS	46.2	32-60 PASS	×1.26
σ_y	41.7	30-50 PASS	×1.39
E	3.0	3-5.5 low	×0.96
ϵ_f	4.2 %	(range 3-7 %)	×1.92
✓ strength inside journal range (k≈1)			